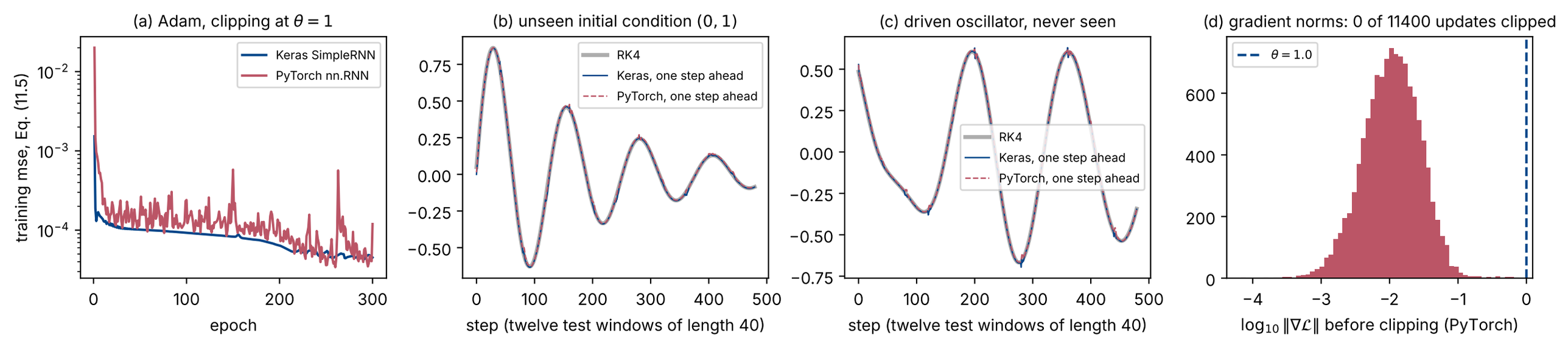
(a) Adam, clipping at $\theta = 1$
training mse, Eq. (11.5)
10^{-2}
10^{-3}
10^{-4}
Keras SimpleRNN
PyTorch nn.RNN
0
100
200
300
epoch
(b) unseen initial condition (0, 1)
0.75
0.50
0.25
0.00
−0.25
−0.50
RK4
Keras, one step ahead
PyTorch, one step ahead
0
100
200
300
400
500
step (twelve test windows of length 40)
(c) driven oscillator, never seen
0.50
0.25
0.00
−0.25
−0.50
−0.75
RK4
Keras, one step ahead
PyTorch, one step ahead
0
100
200
300
400
500
step (twelve test windows of length 40)
(d) gradient norms: 0 of 11400 updates clipped
$\theta = 1.0$
600
400
200
0
−4
−3
−2
−1
0
$\log_{10} \|\nabla \mathcal{L}\|$ before clipping (PyTorch)